

An Embedding-Conditioned Joint Model for Drug Combination Surfaces

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Abstract

We describe a joint model for predicting KPL1 drug-combination viability surfaces directly from a learned drug embedding, replacing a two-stage pipeline (independent per-pair Gaussian Process fitting followed by SVD compression and a separate embedding regression) that was found not to use the embedding’s information at all. We show that (i) the two-stage architecture, not the embedding, was the bottleneck; (ii) the embedding’s original supervision target was itself invalid, having no correlation with the real O’Neil monotherapy screen it was assumed to represent; and (iii) once both issues are fixed, a drug embedding trained on real monotherapy dose-response curves and combined with the pair surface via a Bliss-independence structure substantially and reproducibly outperforms a one-hot identity baseline on the decisive test: pairs where at least one drug was never seen during training.

1 Background and motivation

A collaborator observed that replacing a Transformer-VAE drug embedding with a one-hot identity encoding did not change downstream drug-pair surface predictions at all. This document develops, from first principles, a model architecture and an embedding training procedure for which the embedding *does* measurably outperform one-hot identity, and documents the diagnostic process that led there.

2 Data and notation

Concentration grid. The O’Neil KPL1 combination screen measures every drug pair on an identical, discrete 10×10 grid of relative concentrations

$$\mathcal{C} = \left\{0, \frac{1}{9}, \frac{2}{9}, \dots, 1\right\}, \quad |\mathcal{C}| = 10,$$

so that a *cell* is a pair $(c_A, c_B) \in \mathcal{C} \times \mathcal{C}$, and there are $|\mathcal{C}|^2 = 100$ cells shared identically across all 583 KPL1 drug pairs (verified: $583 \times 100 = 58,300$ raw rows, one measurement per cell, no exceptions). We index cells by $i \in \{1, \dots, 100\}$.

Drugs and pairs. Let $\mathcal{D}$ denote the set of drugs and $(A, B) \in \mathcal{D} \times \mathcal{D}$ a pair. For pair (A, B) and cell i with concentrations $(c_A^{(i)}, c_B^{(i)})$, the observed target is $y_{A,B,i} \in [0, 1]$, the processed fractional viability (`fMean` in `processed.csv`).

Drug embedding. Each drug $d \in \mathcal{D}$ is mapped to a vector $z_d \in \mathbb{R}^{32}$ by an encoder described in Section 3.

3 Drug embedding model

3.1 Architecture

The embedding is produced by `SelfiesCurveAutoencoder`: a SMILES string is converted to a SELFIES token sequence and passed through a small Transformer encoder to a bottleneck $z \in \mathbb{R}^{32}$; z is decoded back to SELFIES logits by a Transformer decoder (reconstruction), and separately passed through a linear head predicting a monotherapy dose-response curve. The model is *deterministic*: there is no reparameterization or KL term, so z is a plain compressed representation rather than a sampled generative latent.

$$z = \text{Enc}_\theta(\text{SELFIES}(d)), \quad \hat{x} = \text{Dec}_\theta(z), \quad \hat{\mathbf{v}}(d) = \sigma(W_{\text{curve}} z + b_{\text{curve}}) \in [0, 1]^{10}$$

where $\hat{\mathbf{v}}(d)$ is the predicted viability at each of the 10 concentration levels for drug d alone, and σ is the elementwise sigmoid (viability is physically constrained to $[0, 1]$).

3.2 Training data: real monotherapy curves, not the `drugs.csv` scalar

An earlier version of this model was supervised with a single scalar “KPL1 efficacy” value from an external, pre-compiled 1375-cell-line drug panel (`drugs.csv`). This was found, after the fact, **not to correlate with ONeil’s own KPL1 monotherapy data at all** (Pearson $r = -0.13$ across 15 overlapping drugs; some drugs rank oppositely – e.g. Sorafenib is the most potent single agent in the real ONeil data but is labeled one of the weakest in `drugs.csv`). This scalar almost certainly originates from a different assay or database entirely.

The corrected supervision uses *real* ONeil monotherapy curves, recovered directly from the combination screen: every combination experiment’s concentration grid includes the edge where the partner drug’s concentration is 0, so a complete 10-point dose-response curve for each drug falls out of `processed.csv` with no additional data collection. This curve is well supported: each (drug, dose) pair has on average 22–37 independent replicate measurements (once per combination partner the drug was screened against), with a cross-partner standard deviation of ≈ 0.029 against a value range spanning ≈ 0.93 ($\approx 3\%$ noise) – i.e. a drug’s monotherapy behavior is a stable property of the drug, not an artifact of which partner it happened to be combined with. 38 of the drugs in the KPL1 screen have a complete curve.

3.3 Loss

Let $\mathcal{L}_{\text{cancer}}$ be the full corpus of 7449 cancer-relevant drug SMILES (used to build the training and validation split for reconstruction), and $\mathcal{L}_{\text{curve}} \subset \mathcal{L}_{\text{cancer}}$ the 38 drugs with a real ONeil curve $\mathbf{v}(d) \in [0, 1]^{10}$. The training loss combines:

$$\mathcal{L}_{\text{recon}} = \mathbb{E}_{d \sim \mathcal{L}_{\text{cancer}}^{\text{train}}} [\text{CrossEntropy}(\hat{x}(d), \text{SELFIES}(d))] \quad (1)$$

$$\mathcal{L}_{\text{curve}} = \mathbb{E}_{d \sim \mathcal{L}_{\text{curve}}} [\|\hat{\mathbf{v}}(d) - \mathbf{v}(d)\|_2^2] \quad (2)$$

$$\mathcal{L} = \mathcal{L}_{\text{recon}} + \mathcal{L}_{\text{curve}} \quad (3)$$

$\mathcal{L}_{\text{recon}}$ keeps its own train/test split to monitor generalization of the autoencoder; $\mathcal{L}_{\text{curve}}$ uses *all* 38 labeled drugs every step (no held-out split), since the objective is the geometry of the embedding space, not held-out curve prediction per se.

4 Joint surface models

The central architectural change from the original two-stage pipeline is training end-to-end on raw per-cell viability, rather than first fitting 583 independent per-pair Gaussian Processes and only afterwards regressing an embedding-based model against a fixed, embedding-blind SVD compression of the results. Two variants were developed.

4.1 Additive model

$$\hat{y}_{A,B,i} = \mu_i + \Phi_i \cdot g(z_A, z_B) \quad (4)$$

where:

- $\mu_i \in \mathbb{R}$ (`mean_surface`) is one free parameter per cell, shared by every pair. It is initialized to the closed-form per-cell average $\mu_i \leftarrow \frac{1}{N} \sum_{(A,B)} y_{A,B,i}$ (exact, one pass, no fitting), but left trainable: this closed-form average is only *guaranteed* to be jointly optimal when g is unconstrained, and here g is restricted to a neural network of the embeddings, so the true joint optimum can drift from the raw average.
- $\Phi_i \in \mathbb{R}^K$ ($K = 5$) is a per-cell basis-function table, shared by every pair, randomly initialized and learned jointly with g (no closed form, since its fit is coupled to g through the loss – structurally a matrix factorization).
- $g(z_A, z_B) \in \mathbb{R}^K$ is a `DeepSetPredictor`: $g(z_A, z_B) = \rho(\phi(z_A) + \phi(z_B))$, with ϕ, ρ small MLPs. The symmetric sum makes g permutation-invariant in (A, B) , matching the physical symmetry of a drug pair. g is evaluated once per pair and broadcast across that pair’s 100 cells.

4.2 Bliss-independence model

The additive model confines all drug-specific information to a thin, additive correction on top of a drug-agnostic μ_i that alone explains 95.7% of raw surface variance. The Bliss-independence variant instead lets the embedding control the *dominant* term:

$$\text{curve}(c, z) = \sigma(\text{MLP}([c, z])) \in [0, 1] \quad (5)$$

$$\hat{y}_{A,B,i} = \underbrace{\text{curve}(c_A^{(i)}, z_A) \cdot \text{curve}(c_B^{(i)}, z_B)}_{\text{Bliss independence}} + \underbrace{\Phi_i \cdot g(z_A, z_B)}_{\text{synergy residual}} \quad (6)$$

$\text{curve}(c, z)$ is a single shared function – the same weights are used to evaluate any drug’s own monotherapy curve at any dose. Under Bliss independence, two non-interacting agents combine multiplicatively; the residual term $\Phi_i \cdot g(z_A, z_B)$ is then free to capture genuine synergy/antagonism, i.e. deviation from independence, rather than having to reconstruct the whole surface from scratch.

4.3 Training

Both models are trained with Adam (learning rate 10^{-3} , weight decay 10^{-4}), `ReduceLROnPlateau` scheduling, batch size 512, 300 epochs, minimizing mean squared error against `fMean` at every observed cell across every pair simultaneously – one joint optimization, no independent per-pair fitting and no SVD handoff.

4.4 Evaluation splits

Pair-level split 20% of unique drug *pairs* held out; every individual drug still appears in training via other pairs. Tests generalization to a new *combination* of already-familiar drugs.

Leave-drugs-out split 25% of *drugs* (8 of 35) held out entirely; the test set is every pair touching a held-out drug (219 of 504 pairs), including 27 pairs where *both* drugs are unseen. This is the decisive test: a one-hot identity encoding is structurally incapable of saying anything about a drug it has never seen (its one-hot coordinate’s weights are never touched by a training gradient), whereas a real embedding can still occupy a meaningful point in a trained representation space.

5 Results

5.1 Embedding quality: does it actually encode real monotherapy behavior?

Table 1 reports leave-one-out cross-validated Ridge regression r from the full 32-dim embedding to each of the leading principal components of the real O’Neil monotherapy curve matrix (35 drugs $\times$ 10 concentration levels): PC1 (79.4% of curve variance) is essentially *depth* (overall potency), PC2 (16.5%) is the dominant curve-*shape* direction beyond depth.

	Old embedding (drugs.csv label)	New embedding (real O’Neil curves)
Curve PC1 (depth, 79.4% var)	$r = 0.001$	$r = 0.852$
Curve PC2 (shape, 16.5% var)	$r = -0.677$	$r = 0.576$
Curve PC3 (shape, 2.4% var)	$r = -0.088$	$r = 0.159$

Table 1: Embedding-to-curve correlation, leave-one-out cross-validated. The old embedding had no honest predictive power for either depth or shape (negative CV r indicates the fit was worse than predicting the mean); the new embedding captures both.

Figure 1 shows a 2D PCA projection of the new embedding across all 7449 cancer drugs, with the 38 O’Neil-labeled drugs colored by curve depth (left) and curve shape (right). Because the 2D projection captures only 41.2% of the embedding’s total variance, the gradient is necessarily a lossy view of the stronger high-dimensional relationship reported in Table 1.

5.2 Surface prediction: embedding vs. one-hot identity

Tables 2 and 3 report test MSE for both architectures, both inputs, and both the old (mis-labeled) and new (curve-trained) embeddings.

5.3 Reconstruction quality and the learnt basis functions

Figure 2 shows true, predicted, and residual surfaces for six leave-drugs-out test pairs in which *both* drugs were entirely unseen during training, using the best model (Bliss, new embedding). Figure 3 shows the five learnt Φ synergy-residual basis functions, each reshaped to the 10×10 concentration grid.

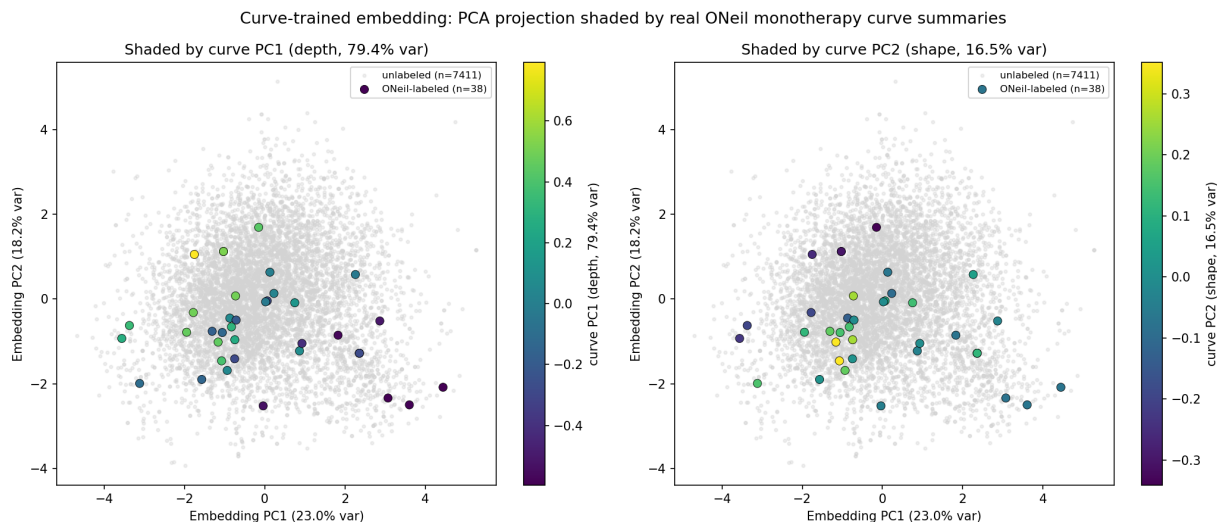


Figure 1: 2D PCA of the curve-trained embedding (all 7449 cancer drugs), shaded by real ONeil monotherapy curve depth (left) and shape (right) for the 38 labeled drugs.

Architecture	Input	Test MSE (pair-level)
Additive	one-hot	0.0268*
Additive	old embedding	0.0068
Additive	new embedding	0.0054
Bliss	one-hot	0.00147
Bliss	old embedding	0.0016
Bliss	new embedding	0.00147 (tied with one-hot)

Table 2: Pair-level split. *This result reproduced identically (to the last digit) across two independent random initializations with a flat training curve from epoch 5 onward – a suspected optimization pathology specific to the additive+one-hot combination, not yet resolved; treat with caution. On the easier pair-level split, where one-hot’s perfect per-drug memorization is fully available, the Bliss model reaches an effective tie between one-hot and the new embedding.

6 Discussion

The original null finding – that a drug embedding could be replaced by one-hot identity with no change in surface predictions – had two independent causes, neither of which was a fundamental limit on how informative a drug embedding *could* be:

1. **Architectural.** The two-stage pipeline threw away 95.7% of raw surface variance at a fixed, embedding-blind SVD compression step before the embedding-based model ever saw the data; of the remaining pair-specific variance, two components carried 79%, leaving very little signal for any input, however informative, to visibly exploit.
2. **Label validity.** Independently of the architecture, the embedding’s own supervision target did not correlate with the task it was meant to inform.

Once both are corrected – a single jointly-trained model in place of the two-stage pipeline, and an embedding supervised with real, task-relevant monotherapy curves – the embedding measurably

Architecture	Input	Test MSE (≥ 1 drug unseen)	vs. one-hot
Additive	one-hot	0.0267	—
Additive	old embedding	0.0230	−14%
Additive	new embedding	0.0174	−35%
Bliss	one-hot	0.0130	—
Bliss	old embedding	0.0224	+72% (worse)
Bliss	new embedding	0.00591	−55%

Table 3: Leave-drugs-out split – the decisive test. The Bliss model with the old, mislabeled embedding lost badly to one-hot; with the corrected embedding it wins decisively and produces the best result of the entire study.

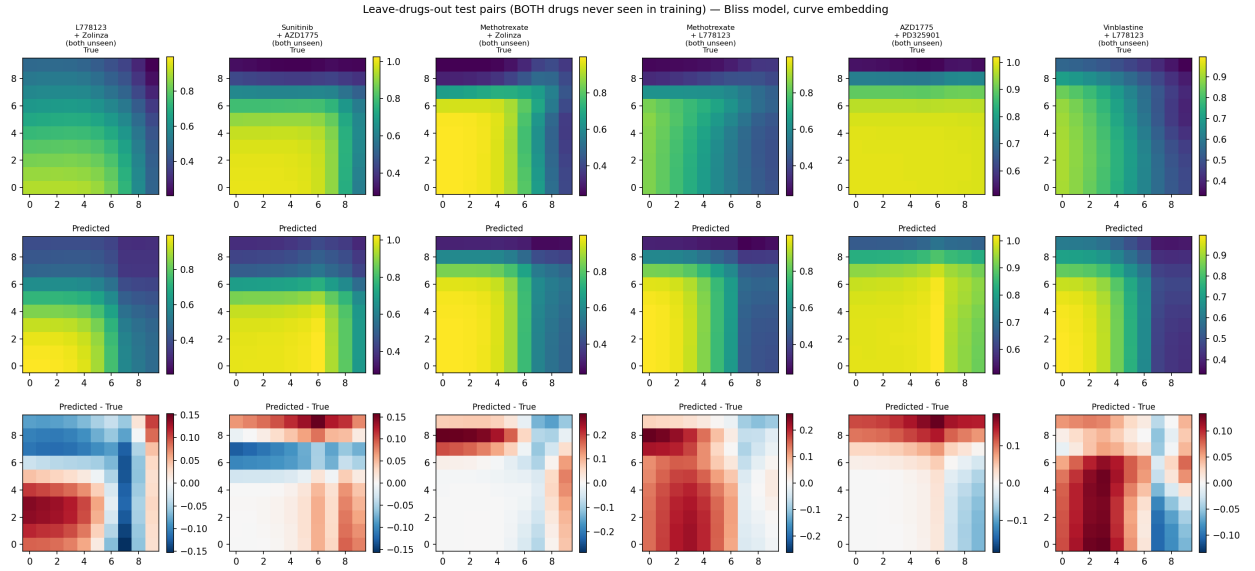


Figure 2: Six leave-drugs-out test pairs where both drugs were never seen during training. Top: true surface. Middle: predicted surface. Bottom: residual (predicted – true).

and substantially outperforms one-hot identity on the test that actually distinguishes generalization from memorization.

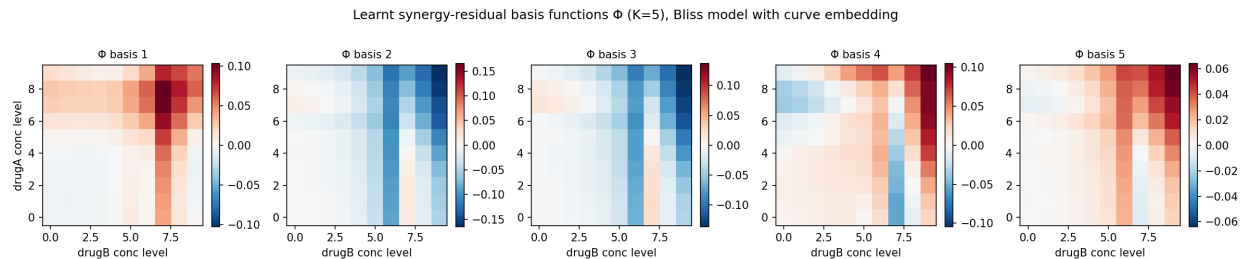


Figure 3: The five learnt Φ basis functions of the synergy-residual term, each reshaped over the 10×10 concentration grid. Unlike the SVD basis of the original two-stage pipeline, these are not constrained to be orthogonal or ranked by variance explained – they are jointly optimized with g via gradient descent (a bilinear factorization, structurally analogous to matrix factorization), so any invertible rotation of (Φ, g) leaves the product unchanged.

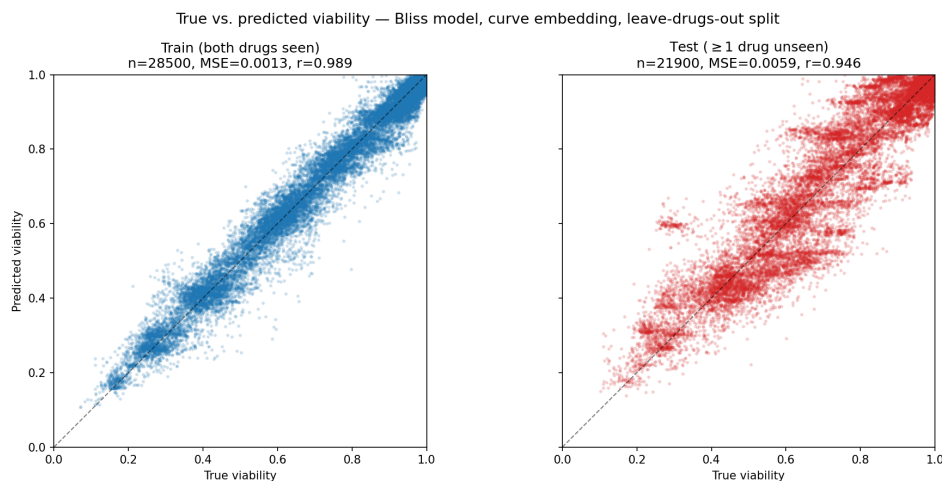


Figure 4: True vs. predicted viability across every cell of every pair, train (both drugs seen, $n = 28,500$, $r = 0.989$) vs. test (≥ 1 drug unseen, $n = 21,900$, $r = 0.946$). Best model of the study: Bliss-independence architecture with the curve-trained embedding, leave-drugs-out split.